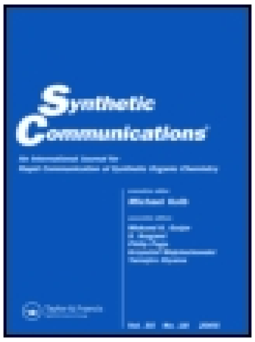

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A New Method for Synthesis of Functionalized Organozinc-Copper(I) Reagents and Their 1,4-Addition Reaction with Enones

Yulai Hu , Jianhua Yu , Shiyan Yang , Jin-Xian Wang & Yuanqi Yin

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A NEW METHOD FOR SYNTHESIS OF FUNCTIONALIZED ORGANOZINC-COPPER(I) REAGENTS AND THEIR 1,4-ADDITION REACTION WITH ENONES

Yulai Hu,[a] Jianhua Yu,[a] Shiyan Yang,[a*] Jin-Xian Wang,[b] Yuanqi Yin[a]

[a]State Key Laboratory for Oxo Synthesis and Selective Oxidation, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, Lanzhou 730000, P.R. China

[b]Department of Chemistry, Northwest Normal University, Lanzhou 730070, P.R. China

Abstract: The transmetalation of functionalized organozinc iodides with $Cu(OAc)_2$ in THF provides a new convenient method for the preparation of functionalized organozinc-copper(I) reagents FG-RCu(OAc)ZnI which ultimately undergo 1,4-addition reaction with enones to give the products in good to excellent yields.

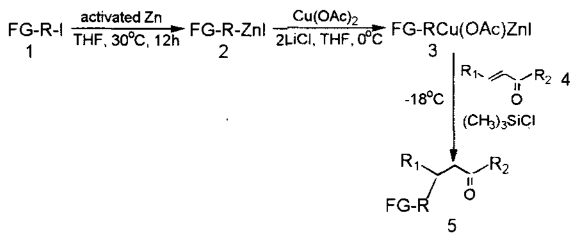
* To whom correspondence should be addressed.

It has been demonstrated that functionalized organozinc reagents are very useful in organic synthesis because of their tolerance of many functionalities, their selectivity and their reactivity towards a wide range of organic electrophiles in the presence of a copper salt like CuCN·2LiCl[1-3]. They can be readily prepared by the insertion of activated zinc into organic iodides[4]or bromides[5]. Because of their inactivity, copper salt must be added in most cases when they react with electrophiles. As we know, CuCN has almost been the only copper salt since it was first used in 1988[1-7]. Because of the toxicity of CuCN, we tried to find another way to carry out the transmetalation instead of CuCN. In our experiment, we used the mixture of $Cu(OAc)_2$ and LiCl, and found that it could be successfully used in the synthesis of organozinc-copper reagents, which could react with enones under very mild conditions to afford 1,4-addition products in good to excellent yields. We now report a new method for the synthesis of the organozinc-copper reagents and their 1,4-addition reaction with enones(Scheme I):

The organozinc reagents were obtained following Knochel's procedure[4], and then added into a solution of $Cu(OAc)_2$ and LiCl in THF at 0°C to give the organozinc-copper reagents. XPS shows that copper exists as Cu^+ (Cu $2p^3$ 932.13ev) and zinc exists as Zn^{2+} (Zn $2p^3$ 1022.01ev) in the organozinc-copper reagents. Thus, we carried out the reaction with CuOAc instead of $Cu(OAc)_2$, and the same result was obtained and shown in Table I, entry 2. So we tentatively suggest that $Cu(OAc)_2$ is reduced to CuOAc after the addition of FG-RZnI and then FG-RCu(OAc)ZnI is formed through transmetalation of FG-RZnI with

CuOAc. Introduction of an enone leads to 1,4-delivery of the functionalized group FG-R- (Scheme I).

Scheme I



2a R= n-C_5H_{11}-

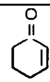
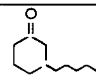
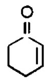
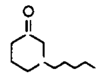
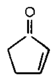
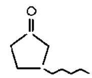
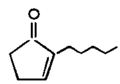
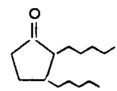
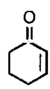
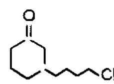
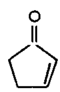
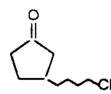
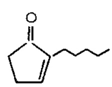
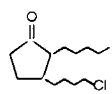
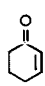
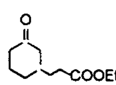
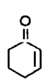
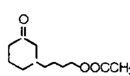
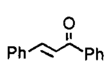
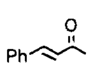
2b R= $ClCH_2CH_2CH_2CH_2$-

2c R= $CH_3CH_2OOCCH_2CH_2$-

2d R= $CH_3COOCH_2CH_2CH_2CH_2$-

In the 1,4-addition reaction of FG-RCu(OAc)ZnI with enones, the addition of LiCl and $(CH_3)_3SiCl$ is necessary. Without LiCl, the addition of pentylzinc iodide to$Cu(OAc)_2$ in THF leads only to very low yield of 3-pentylcyclohexanone because of the insolubility of $Cu(OAc)_2$ in THF, and in the absence of $(CH_3)_3SiCl$, 1,4-addition product 3-pentylcyclohexanone can not be obtained. However, temperature has little effect on the reaction when it is below -18°C. For example, the yield of 3-pentylcyclohexanone is 78% or 82% when pentylzinc iodide reacts with 2- cyclohexenone at -70°C or -18°C, respectively. The structure of enones has a great effect on the reaction. When β-position of enones has an aryl group, the reaction can not be conducted(Table I, Entry 10, 11).

Table I: 1,4-Addition reaction of FG-RZnI mediated by $Cu(OAc)_2{\cdot}2LiCl$

Entry	FG-R-ZnI	Enone	Product	Yield(%)
1	2a			82
2	2a			78
3	2a			85
4	2a			74
5	2b			78
6	2b			78
7	2b			71
8	2c			75
9	2d			68
10	2a			0
11	2a			0

In summary, we have disclose a new transmetalation process for converting functionalized organozinc halides to reactive organozinc-copper reagents which can undergo 1,4-addition reaction with enones efficiently under very mild condition to give the products in good to excellent yields. The methodology uses $Cu(OAc)_2$ as the copper salt instead of CuCN. Thus it is a more convient and "environmentally friendly" approach. Further studies concerning the mechanism and the reactions of these organometallics towards other electrophiles are under way in our laboratory.

Experimental

All manipulation were performed under an atmosphere of dry nitrogen gas. Solvents were refluxed over sodium before use. Infrared spectra were run on 10DX-FTIR spectrometer as thin films between KBr windows. Mass spectra were recorded on QP-1000GC-MS spectrometer. NMR spectra were obtained in $CDCl_3$ on AC-80 spectrometer or FT-400 spectrometer. The zinc dust, cupric acetate, lithium chloride, chlorotrimethylsilane and n-pentyl iodide were purchased. The following starting materials were prepared according to the literature: 2-cyclopetenone[8], 2-cyclohexenone[8], 2-pentyl-2-cyclopentenone[9], 1-chloro-4-iodobutane[10]. The ethyl 3-iodopropionate and 4-iodobutyl acetate were prepared from the corresponding chlorides and sodium iodide[11].

General procedure for the preparation of FG-RCu(OAc)ZnI and their reaction with enones:

Under nitrogen atmosphere, the mixture of zinc(0.026 mol), 1,2-dibromoethane(0.09ml) and THF(2ml) in a three-neck flask was heated to 60-70°C for a minute and then cooled to room temperature.Chlorotrimethylsilane(0.1ml) was added, and the mixture was stirred at room temperature for 15 minutes. A solution of FG-R-I(0.025mol) in THF(10ml) was then added, and the mixture was stirred 12h at 35-40°C. The resulting solution of FG-RZnI in THF was ready to use.

Anhydrous $Cu(OAc)_2$(0.013mol), LiCl(0.026mol) and THF(15ml) were put into another three-neck flask charged with nitrogen gas. The mixture was stirred at room temperature for 30 minutes and cooled to -18°C with salt-ice bath. The solution of n-C_5H_{11}-ZnI in THF was added dropwise into the flask. After the end of the addition, the mixture was stirred 10 minutes at -18°C. A solution of enone(0.012mol) and chlorotrimethylsilane(3ml) in Et_2O(10ml)was added in 20 minutes. The reaction mixture was stirred at -18°C ~ -10°C for 1h and then allowed to warm to room temperature. After stirring for 5h, the mixture was quenched with saturated aqueous NH_4Cl and extracted with Et_2O, and the combined organic layer was dried over $MgSO_4$, and concentrated. The product was isolated from the crude reaction mixture by chromatography on silica gel(diethyl ether/petrolum ether).

5a: Liquid, υ_{max}(neat)/cm^{-1} 2955,2988, 1714, 1458; δ_H(80MHz;$CDCl_3$) 2.38-2.02(m, 4H), 1.86-1.28(m,13H), 0.88(m,3H); EI-MS m/z 168(M^+,56%), 153(6), 139(11), 125(100), 111(24), 97(98), 83(99), 81(60),55(98).

5b: Liquid, υ_{max}(neat)/cm^{-1} 2959, 2926, 2856, 1743,1462; δ_H(400MHz;$CDCl_3$) 2.3-2.0(m, 4H), 1.71(q,2H), 1.44-1.23(m, 9H), 0.82(t, 3H); EI-MS m/z 154(M^+, 2%), 126(4), 83(100),70(20), 55(68), 28(42).

5c: Liquid, υ_{max}(neat)/cm^{-1} 2957, 2928, 1739, 1462; δ_H(80MHz;$CDCl_3$) 2.25-2.01(m, 3H), 1.69-1.29(m, 19H) 0.90-0.87(m, 6H); EI-MS m/z 224(M^+, 21%) 153(100), 125(42), 96(13), 82(50), 54(60).

5d: Liquid, υ_{max}(neat)/cm^{-1} 2953, 2930, 2860, 1712, 680; δ_H(80MHz;$CDCl_3$) 3.54(t, 2H), 2.33-2.03(m, 4H), 1.85-1.37(m, 11H); EI-MS m/z 188(M^+, 5%), 145(22), 97(100), 69(16), 55(52).

5e: Liquid, υ_{max}(neat)/cm^{-1} 2957, 2933, 2864, 1739,650; δ_H(80MHz;$CDCl_3$) 3.54(t, 2H), 2.29-2.08(m, 3H), 1.89-1.47(m, 10H); EI-MS m/z 174(M^+, 36%), 145(69), 139(16), 111(6), 97(32), 83(100), 55(97).

5f: Liquid, υ_{max}(neat)/cm^{-1} 2955, 2930, 2860, 1739, 651; δ_H(80MHz;$CDCl_3$) 3.57(t, 2H), 2.27-2.04(m, 3H), 1.91-1.27(m, 17H), 0.88(t, 3H); EI-MS m/z 244(M^+, 3%), 189(2), 174(8), 153(10), 139(2), 97(2), 83(100).

5g: Liquid, υ_{max}(neat)/cm^{-1} 2980, 2933, 2866, 1732, 1711; δ_H(80MHz;$CDCl_3$) 4.10(q, J=7.08Hz,2H), 2.40-2.08(m, 6H), 2.01-1.38(m, 7H), 1.2(t, J=7.09Hz, 3H); EI-MS m/z 198(M^+, 8%), 153(47), 125(6), 111(41), 97(100). 69(36).

5h: Liquid, υ_{max}(neat)/cm^{-1} 2953, 2807, 1739, 1685, 1238, 1045; δ_H(80MHz;$CDCl_3$) 4.01(t, 2H), 2.23- 2.08(m, 4H), 2.02(s, 3H), 1.83-1.08(m, 11H); EI-MS m/z 212(M^+, 49%), 197(6), 153(4), 141(69), 125(5).

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